Freight Rate Prediction Assessment Report

Spotter Machine Learning Engineer take-home

Objective: predict **posted_rate** ($) for truckload shipments using route, equipment, weight, calendar, and market-signal features.

1. Data overview & quality

train_test.csv: 48,000 labeled loads from 2025-01-01 to 2025-10-31. **validation.csv**: 12,000 unlabeled loads from 2025-11-01 to 2025-12-31. Target column is **posted_rate** (present only in train).

Issues found and how they were handled:

- Missing **weight** (~0.6%) and **market_index** (~0.8%) → median imputation fitted on the training fold only.
- Right-skewed rates → model **log1p(rate_per_mile)**, then multiply by distance (keeps economics sensible).
- December chart inputs omit lat/lon and market fields → city coordinates looked up from history; daily market_index / quote_signal means taken from the validation period (features only).
- Equipment is categorical (Dry Van / Reefer / Flatbed) with different typical $/mile.

2. Train / validation split

Because the official validation window is later in time, I used a **time-based holdout** instead of a random split: train through **2025-09-30** (43,147 rows), evaluate on **2025-10-01–2025-10-31** (4,853 rows). This tests temporal generalization the same way the Nov–Dec scoring set will.

3. Features & model

Features include distance and weight transforms, haversine vs reported distance gap, pickup/delivery/route categoricals, equipment, market × distance interactions, and cyclical day-of-year / day-of-week encodings for seasonality outside the training calendar window.

Model: **LightGBMRegressor predicting log1p(rate_per_mile)** (LightGBM). After holdout tuning / early stopping (best iteration ≈ 77), the model was **refit on all 48,000** training rows before scoring validation and December rows.

4. Holdout results (October 2025)

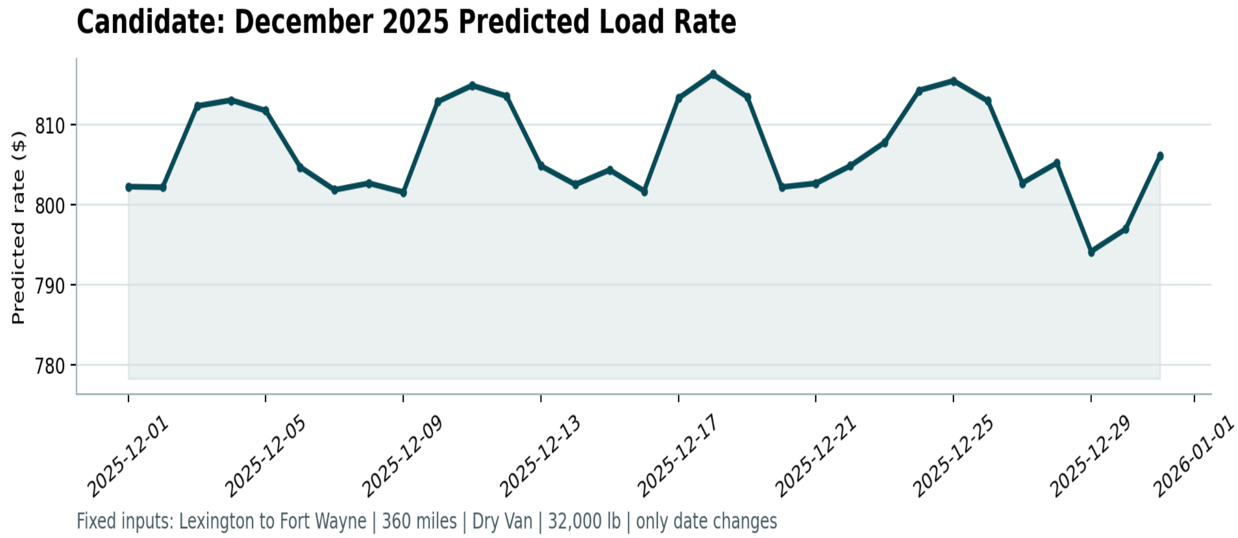
LightGBM — MAE **$129.83**, RMSE **$653.15**, MAPE **5.84%**, R^2 **0.817**.

Distance × median-$/mile baseline — MAE **$253.85**, MAPE **11.81%**, R^2 **0.793**. The model roughly halves MAE versus the naive mileage baseline.

Top features by split importance: quote_signal, delivery, pickup, weight_per_mile, market_x_quote, weight, quote_x_distance, dayofyear.

5. December fixed-lane chart

Lane held fixed: Lexington → Fort Wayne, 360 miles, Dry Van, 32,000 lb. Only the date (and imputed daily market signals) change. Predicted rates move modestly with market/calendar conditions while staying near historical Dry Van rates for this corridor.



6. How to reproduce

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
cd src && python train_predict.py && cd ..
python score.py --predictions validation_predictions.csv \
  --december-predictions data/december_chart_inputs.csv
```

Deliverables: validation_predictions.csv, filled december_chart_inputs.csv, this report, source under src/, and scorer_results/candidate_december.png.